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A REPORT ON MAJOR PROJECT PHASE-II WORK

IoT BASED SCHOOL BUS MANAGEMENT USING RFID

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Project Associates

ANURADHA P	4BD22EC011
SANJAY S SHESHAGIRI	4BD22EC079
SHUBHASHREE B M	4BD22EC121

Dr. KIRAN KUMAR G H

M. Tech., Ph. D., MISTE, MIETE, MIE

Project Guide

Dr. G. S. SUNITHA

M. Tech. (DEAC), Ph. D., MISTE, FIETE, FIE

Program Coordinator

Dr. H.B. ARAVIND

M.E. (Env. Eng.), Ph. D., FIE, MISTE, FIEE

Principal



Bapuji Educational Association®

Bapuji Institute of Engineering and Technology

Davangere-577 004

Department of Electronics & Communication Engineering

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BAPUJI INSTITUTE OF ENGINEERING AND TECHNOLOGY

DAVANGERE, KARNATAKA – 577004



DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING

Certificate

Certified that the Project work entitled **IoT Based School Bus Management Using RFID** carried out by Ms. Anuradha P, USN 4BD22EC011, Mr. Sanjay S Sheshagiri, USN 4BD22EC079, Ms. Shubhashree B M, USN 4BD22EC121, bonafide student of this institution in partial fulfillment for the award of degree of **Bachelor of Engineering in Electronics & Communication** by **Visvesvaraya Technological University**, Belagavi during the academic year **2025-26**. It is certified that all corrections / suggestions indicated for internal assessment have been incorporated in the report deposited in the departmental library. The project report has been approved as it satisfies the academic requirements in respect of Project work prescribed for the said degree.

Dr. Kiran Kumar G H

M. Tech., Ph. D., MISTE, MIETE, MIE

Project Guide

Dr. G.S. Sunitha

M.Tech. (DEAC), Ph.D., MISTE, FIETE., FIE

Professor & Head

Dr. H.B. Aravind

M.E., (Env. Engg.) Ph.D., FIE., MISTE., FIEE

Principal

External Viva-voce

Name of the Examiners

Signature with Date

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2.

.....

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ABSTRACT

The IoT-Based School Bus Management System Using RFID is designed to enhance student safety, monitoring accuracy, and transportation efficiency in school bus operations. The traditional method of bus tracking is prone to errors, delays, and security risks, such as unauthorized boarding and lack of real-time visibility for parents and school authorities. To overcome these issues, the proposed system integrates RFID authentication, GPS-based live tracking, IoT cloud monitoring, and an automated door-locking mechanism controlled by a solenoid lock.

When a student scans their RFID card at the bus entrance, the system verifies their identity through the ESP32 microcontroller and displays the authentication status on the LCD. Valid entries trigger the automatic unlocking of the door, while invalid attempts activate a buzzer and alert indication. Simultaneously, the GPS module updates the real-time location of the bus to the Arduino IoT Cloud, enabling parents and school administrators to track the bus movement and receive student boarding logs through the mobile application. This enhances transparency, improves response time during emergencies, and provides a reliable digital attendance record.

The implemented system offers a secure, automated, and scalable solution for modern school transportation by ensuring controlled entry, continuous monitoring, and efficient data management through IoT integration. It serves as a practical model for developing smarter and safer school transport infrastructures.

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Project Associate

Anuradha P

Sanjay S Sheshagiri

Shubhashree B M

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CHAPTER 1

INTRODUCTION

In most educational institutions, school bus transportation is managed using conventional methods that rely heavily on manual supervision by drivers and conductors. Students board the bus based on visual recognition, and there is no automated mechanism to verify whether the person entering the bus is an authorized student. The responsibility of monitoring student safety, door operation, and route management entirely depends on human observation. Communication between the bus, school authorities, and parents is usually indirect and delayed, often relying on phone calls or verbal updates. As a result, the traditional system operates without technological assistance and lacks real-time visibility into bus operations.

In the conventional system, bus door operation is handled manually by the driver or conductor. This process may lead to unsafe conditions if the door is not closed properly while the bus is in motion. Additionally, there is no systematic method to track the real-time location of the bus. Parents waiting at bus stops have no clear information regarding arrival time, delays, or route deviations, which often causes anxiety and inconvenience. School authorities also face difficulty in monitoring multiple buses simultaneously, especially during peak hours or emergency situations.

Although manual supervision has been used for many years, it becomes less effective as the number of students and buses increases. These limitations indicate that traditional school bus management systems are not sufficient to meet modern safety and monitoring requirements.

The conventional approach presents several significant drawbacks. One of the major issues is unauthorized boarding, where individuals other than registered students may enter the bus without being noticed. Since there is no identity verification mechanism, the system depends entirely on human judgment, which is prone to error. Another major drawback is the absence of real-time location tracking, which makes it difficult for parents and school authorities to know the exact position of the bus at any given time. This lack of visibility reduces transparency and increases uncertainty.

Another critical limitation is manual door control, which may result in improper locking during transit. If the door is not secured correctly, it can pose serious safety risks to students. Additionally, the traditional system does not provide automated alerts or notifications in case of emergencies, delays, or abnormal events. Information is usually communicated late, reducing the effectiveness of

response measures. Overall, the conventional method lacks automation, real-time monitoring, and intelligent decision-making, making it less reliable in ensuring student safety.

To overcome these drawbacks, the proposed IoT-Based School Bus Management System Using RFID introduces an automated, technology-driven solution that enhances safety, reliability, and transparency. The system integrates RFID-based authentication, real-time GPS tracking, automated door control, and IoT cloud monitoring to address the limitations of traditional methods. By incorporating embedded systems and wireless communication, the proposed solution reduces dependency on manual supervision and minimizes human error.

In the project, each authorized student is provided with an RFID card containing a unique identification number. When the RFID card is scanned, the system verifies the identity using the ESP32 microcontroller. Only after successful authentication does the system activate the relay-controlled solenoid lock to open the bus door. This ensures that only authorized students are allowed to enter the bus, effectively preventing unauthorized boarding. For invalid cards, the system keeps the door locked and triggers alert mechanisms.

The project also incorporates a GPS module to continuously track the real-time location of the bus. The location data is transmitted to an IoT cloud platform through the ESP32's Wi-Fi connectivity. Parents and school authorities can monitor the live bus location using a mobile application, enabling better planning and reducing uncertainty during transit. Real-time tracking improves transparency and allows timely action in case of route delays or emergencies.

To further enhance safety and user interaction, the system includes an LCD display, LED indicators, and a buzzer. The LCD provides real-time status messages such as authentication results and system notifications, while LEDs and buzzer alerts provide immediate visual and audio feedback during valid or invalid access attempts. Automated door locking ensures that the bus door remains securely closed while the vehicle is in motion.

Overall, the IoT-based system effectively addresses the key drawbacks of the traditional school bus management approach by introducing automation, real-time monitoring, and intelligent control. By integrating RFID authentication, GPS tracking, cloud-based monitoring, and automated door operation, the system significantly enhances student safety and operational efficiency. This solution demonstrates how modern IoT technologies can be practically applied to transform conventional transportation systems into smart and secure solutions suitable for real-world deployment.

CHAPTER 2

LITERATURE SURVEY

This chapter presents a review of relevant research works related to RFID-based authentication, GPS tracking, and IoT-enabled transportation systems. The literature survey helps in understanding current methodologies, technologies used, and their limitations. Based on this analysis, the need for an improved and integrated IoT-based school bus management system is justified.

1. IoT-Based School Bus and Student Monitoring System Using RFID and GSRM Technologies.

Rashmi Ranjan Das, Christina Josephine Malathi A., Aarthi M, M. Leeban Moses, Deepika Rani Sona (2024)

In this paper by Rashmi Ranjan Das et al., the authors design a smart school bus monitoring system using RFID, GSM, GPS, and IoT technologies. The aim is to provide real-time safety updates and location tracking to both parents and school authorities. Each student carries an RFID tag, which is scanned as they enter or exit the bus. The system uses a GPS module to track the bus's exact location and a GSM module to send alerts and notifications to the school administration or parents via SMS. The system is built on an Arduino UNO platform, which collects the data from sensors and uploads it to ThingSpeak cloud for visualization. The authors highlight how this system improves accountability and reduces parental anxiety by giving them access to their child's boarding and location information.

2. Digitalization of College Bus Transportation System

Nayana Shaji, Alex Issac, Josmy George, Anugraha Saji, Sebin Joseph (2023)

This paper by Nayana Shaji and team focuses on transforming traditional bus management systems into digital, automated, and IoT-enabled frameworks. The main objective is to reduce manual errors in attendance marking, tracking student travel, and fare calculation. The system integrates RFID, GPS, and NodeMCU (ESP32) to achieve complete digitalization. Each student is issued an RFID card, which is scanned when boarding or exiting the bus. The ESP32 microcontroller acts as the central processing unit, handling both RFID and GPS modules and ensuring data is uploaded to the cloud server through Wi-Fi. The system also calculates travel distance to automate fare management, eliminating the need for manual ticketing. The paper emphasizes that this digital system increases operational efficiency and ensures transparency.

3. College bus security and student monitoring system

B.Nithin, V.Vinay Kumar, A. Sathwika, M. Sai Vineeth Reddy (2022)

This paper by B. Nithin et al. presents a real-time school and college bus tracking system focused on enhancing student safety. The system uses RFID for identification, GPS for location tracking, and Wi-Fi connectivity for data transfer. It is implemented using an Arduino Mega microcontroller, which integrates data from the RFID reader and GPS module. The processed data is uploaded to a web server that can be accessed by both school administrators and parents via a mobile application. The authors also included additional safety features like an emergency switch, buzzer alerts, and two-way communication between driver and school authorities.

4. RFID Based College Bus Management System

M. Malleswari, M.Koteswara Rao, K.V.Supriya, K.Pavan Krishna, B.Ravi Teja (2018)

In this paper, M. Malleswari et al. proposed a bus management system that combines RFID, GPS, and GSM technologies to track student movement and automate attendance. The authors focused on improving efficiency in college transportation by replacing manual attendance records with a fully automated system. Each student's RFID card is linked to their identity, and scanning the card on entry or exit updates the student's attendance in the system. The GPS module provides real-time location of the bus, while the GSM module sends this information to the college's main server. The Atmega328 microcontroller acts as the core processing unit, handling RFID input, GPS data, and GSM communication.

5. Bus Tracking and Monitoring Using RFID

Pravin A. Kamble, Rambabu A. Vatti(2017)

This paper by Pravin A. Kamble and Rambabu Vatti presents an IoT-based bus tracking and monitoring system that uses RFID technology and ThingSpeak cloud integration to ensure passenger safety and real-time monitoring. The main goal of this system is to improve public transportation by giving authorities and passengers accurate information about bus location, passenger count, and movement. The system uses RFID tags to identify passengers, while infrared (IR) sensors detect the number of people boarding and leaving the bus. The Arduino microcontroller processes this data and uploads it to the ThingSpeak IoT cloud, where real-time bus location and passenger details can be monitored through a web dashboard.

Table 2.1 Summary of literature survey

Ref. Paper Author/Year	Objective	Parameter Evaluated	Adaptive Techniques	Drawbacks	Improvement
1. Authors: Rashmi Ranjan Das et al. (2024)	To ensure student safety through RFID authentication, GPS tracking, and GSM alerts.	Accuracy of attendance logs, live GPS tracking, emergency SMS reliability.	Arduino UNO, RFID, GPS, GSM, ThingSpeak cloud.	GSM alert delays and limited RFID range.	Migration to IoT cloud with faster communication and mobile app-based parental monitoring.
2. Authors: Nayana Shaji et al. (2023)	To automate college bus ticketing, attendance, and tracking through IoT integration.	Trip distance via GPS, fare deduction logs, RFID card authentication.	NodeMCU (ESP32), RFID, GPS Neo-6M, Cloud connectivity.	Internet dependency and lack of data encryption for student details.	Integration of biometric or facial authentication; AI-based route prediction for improved efficiency.
3. Authors: B. Nithin et al. (2022)	To develop a real-time bus tracking and student monitoring system using IoT.	Real-time GPS feed, student entry/exit timestamps, message alerts.	Arduino Mega, RFID, GPS, GSM/Wi-Fi integration.	Lacks biometric validation; RFID misuse possible.	Use of ESP32 with Arduino Cloud and face recognition to enhance security.
4. Authors: M. Malleswari et al. (2018)	To automate student attendance and bus tracking using RFID and GSM modules.	RFID logging, GPS-based location, data transmission via GSM to a local server.	Atmega328 microcontroller, RFID, GSM, GPS, LCD module.	High delay in GSM-based data transfer, no real-time visualization.	Upgrade to IoT-based dashboards using ESP32 for real-time cloud access.
5. Authors: Pravin A. Kamble, Rambabu Vatti (2017)	To design an IoT-enabled system for real-time bus tracking and passenger identification using RFID and ThingSpeak cloud.	This work enables real-time bus tracking and passenger identification using RFID and IoT, but has limited RFID range and relies on stable network connectivity.	Arduino UNO, RFID reader, IR sensors, ThingSpeak IoT platform.	Dependent on Wi-Fi network for data upload; limited RFID range.	Development of a mobile application and stronger communication protocols for reliable data transfer.

CHAPTER 3

PROBLEM IDENTIFICATION

Based on the insights gained from the literature survey, several practical limitations and unresolved issues in existing school bus management systems were identified. Most conventional systems lack secure access control, real-time monitoring, and effective safety mechanisms. These shortcomings highlight the need for a reliable, automated, and IoT-enabled solution. Hence, this chapter focuses on clearly defining the problems that the proposed system aims to address.

In conventional school bus systems, there is no real-time student tracking, making it difficult for parents and schools to ensure safety during transit. Manually checking student presence is time-consuming and lacks reliability. Additionally, the lack of an authentication mechanism allows unauthorized access, posing serious security risks.

CHAPTER 4

OBJECTIVES

After identifying the key issues in current school bus systems, this chapter outlines the objectives of the proposed project. The objectives are formulated to directly address the identified problems such as unauthorized access, lack of live tracking, and poor monitoring. Each objective focuses on improving safety, transparency, and automation using modern IoT technologies. These objectives guide the design and implementation of the proposed system.

- To develop a smart school bus management system that ensures student security using RFID authentication, real-time GPS tracking, automated door locking, and IoT integration for monitoring.
- To provide a user-friendly interface for parents and school administrators to monitor bus location and student entry logs.

CHAPTER 5

METHODOLOGY

The overall working principle of the proposed system can be understood from the block diagram shown in Figure 5.1.1. The methodology of this project explains how each of the objectives are achieved through a structured design and implementation approach. Based on the objectives, a detailed block diagram is presented to illustrate the interaction between key modules such as RFID, GPS, ESP32, relay, and IoT Cloud. The following sections describe how these components were integrated to create a functional, automated, and secure school bus management system.

5.1 Block Diagram

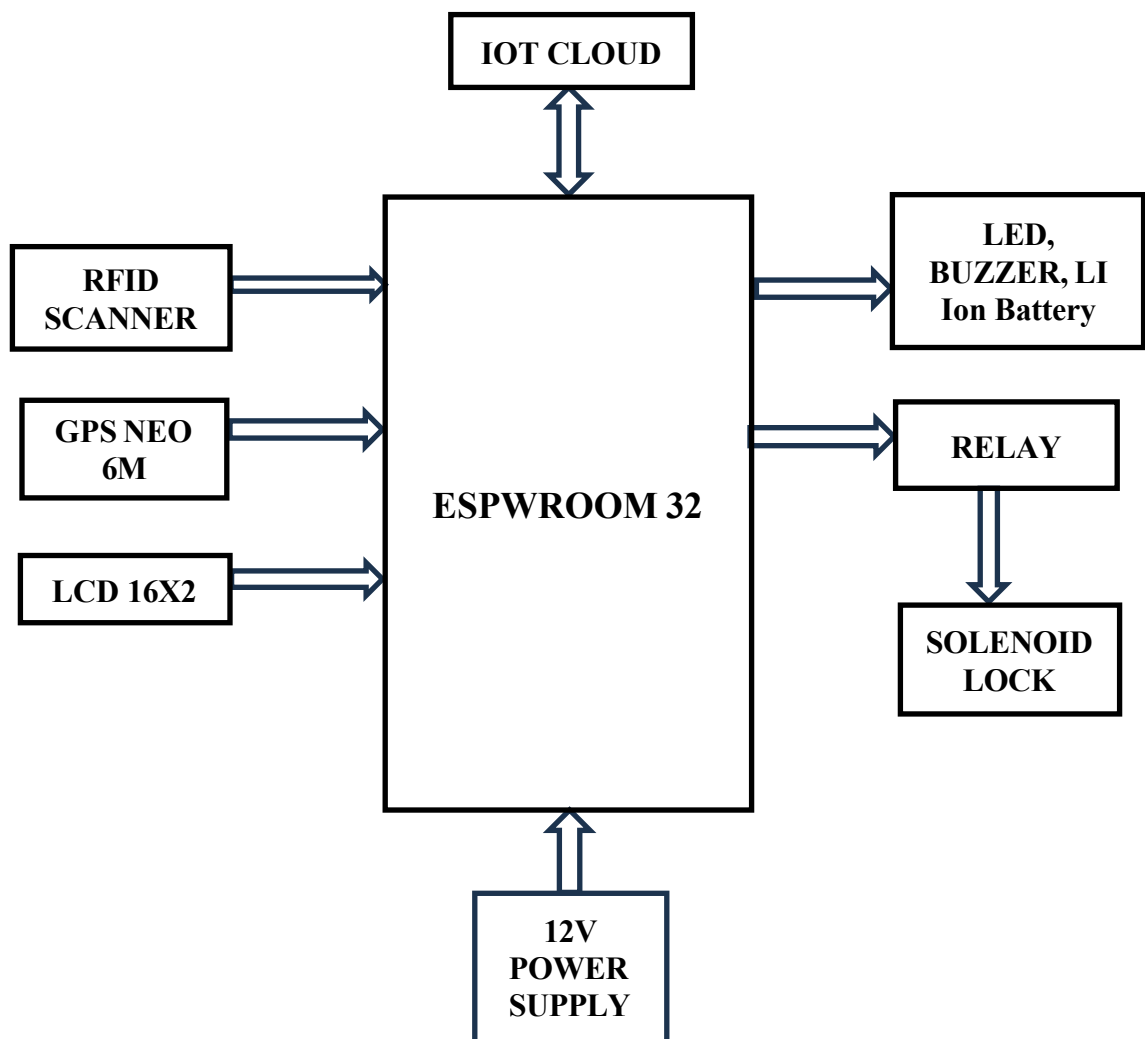


Figure 5.1.1 Block diagram of IoT Based School Bus Management using RFID

The IoT-Based School Bus Management System Using RFID is centered around the ESP32-WROOM32 microcontroller, which acts as the main processing and communication unit. The RFID scanner is interfaced with the ESP32 to authenticate students when they present their RFID card. Once a card is scanned, the ESP32 processes the unique ID number and determines whether the student is authorized to enter the bus.

At the same time, the LCD 16×2 display provides real-time feedback by showing messages such as “Valid ID” or “Invalid ID”. This ensures immediate visual communication between the system and the user.

To enhance safety and automated control, the ESP32 is connected to a relay module that drives the solenoid lock. When a valid RFID card is detected, the ESP32 signals the relay to activate the solenoid lock, allowing the door to open. In the case of an invalid scan, the door remains locked and the buzzer and LED indicators alert the user. The LED, buzzer, and Li-ion battery work together to ensure clear signaling and stable power. The system is also supplied power through a regulated 12V power module which distributes appropriate voltage levels to each component, ensuring reliable and safe operation.

For real-time bus tracking, the GPS Neo-6M module continuously sends latitude and longitude data to the ESP32 using UART communication. The ESP32 then uploads both the GPS information and RFID entry logs to the IoT Cloud. This allows parents and school authorities to remotely monitor the bus location and student entry status through the IoT dashboard or mobile app. The integration of all these modules ensures that the system provides secure boarding, real-time tracking, and remote monitoring, making school transportation safer and more transparent.

CHAPTER 6

HARDWARE DESCRIPTION

The Methodology chapter explained the overall working principle and data flow of the proposed system using block and flow diagrams. To implement this methodology in real time, appropriate hardware components are required. This chapter provides a detailed description of the hardware modules used in the project, including the ESP32, RFID reader, GPS module, relay, solenoid lock, LCD, and power supply. Each component is selected to support the system functionality efficiently and reliably.

6.1 ESP WROOM 32

Figure 6.1.1 shows the ESP32 Microcontroller

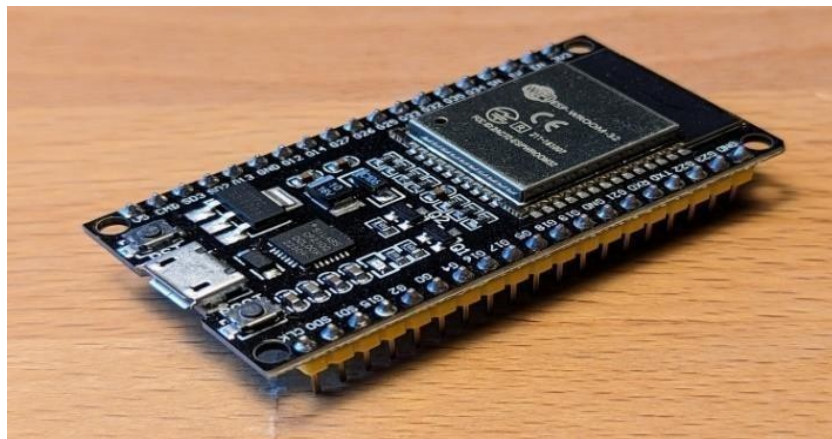


Figure 6.1.1 ESP WROOM 32

Role-Central microcontroller that controls, processes, and manages communication between all modules.

Explanation -The ESP WROOM 32 is a powerful Wi-Fi and Bluetooth-enabled microcontroller used to process RFID data, control outputs like solenoid locks, and communicate with cloud services. It receives input from the RFID, GPS, and sensors, processes it, and triggers appropriate outputs including locking/unlocking doors, displaying messages, and sending cloud updates.

Specifications

- Dual-core Xtensa 32-bit LX6 microprocessor
- Clock speed: up to 240 MHz
- Flash Memory: 4 MB

6.2 GPS Neo 6M

Figure 6.2.1 shows the GPS Neo 6M and Interface shown in figure 6.2.2



Figure 6.2.1 GPS Neo 6M

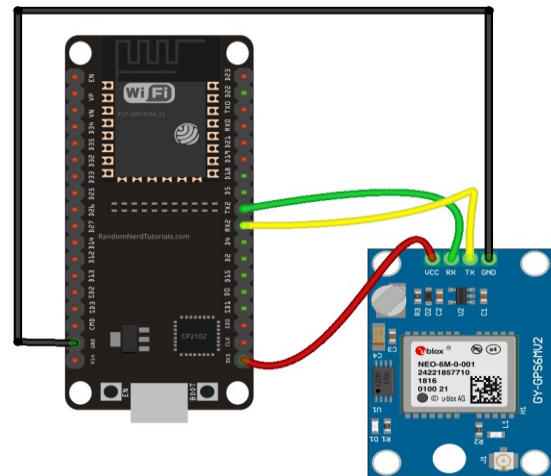


Figure 6.2.2 GPS Neo 6M Interface

Role- The GPS Neo-6M module is responsible for providing the real-time geographical location of the school bus. It continuously sends latitude and longitude coordinates to the ESP32 microcontroller, enabling live bus tracking through the IoT cloud.

Explanation- The NEO-6M is a high-performance GPS receiver that communicates with the ESP32 through UART (TX–RX). Once powered, it connects with GPS satellites and retrieves position data in the form of NMEA sentences. The ESP32 processes this data and uploads it to the Arduino IoT Cloud, allowing parents and school authorities to monitor the bus movement in real time. Its fast satellite locking, stable signal reception, and accurate coordinate output make it highly suitable for smart transportation systems.

Specifications

- Operating Voltage: 3.3V – 5V
- Communication: UART (TX/RX)
- Position Accuracy: ~2.5 meters
- Update Rate: 1 Hz (default)
- Antenna Type: Ceramic patch antenna
- Supported Data Format: NMEA
- LED Indicator: Fix LED blinks when satellite lock is achieved

6.3 Relay module

Figure 6.3.1 shows the Single Channel Relay Module and Interface shown in figure 6.3.2

(1 Channel Relay Module) 5V DC 10A Expansion Board

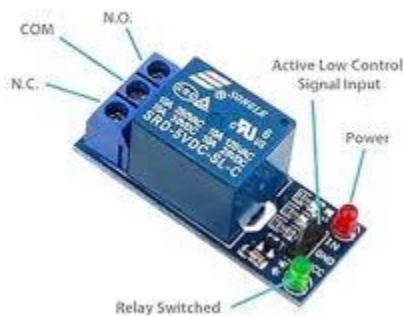


Figure 6.3.1 Single Channel Relay Module

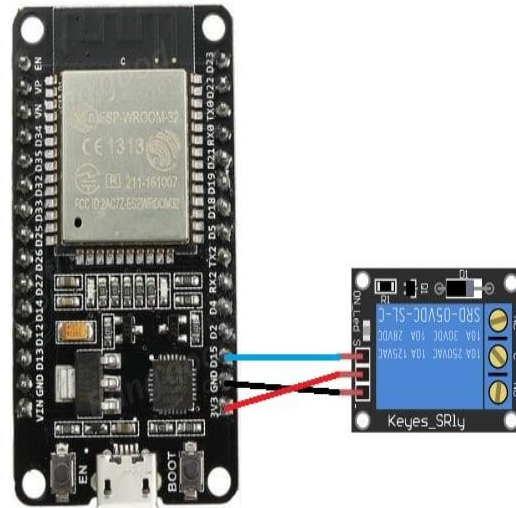


Figure 6.3.2 Relay Module Interface

Role- The 1-channel relay module acts as an electrical switch that allows the ESP32 microcontroller to control high-power devices, such as the solenoid lock used in our project. Since the ESP32 cannot directly handle high voltage/current loads, the relay provides isolation and safe switching. In this system, it enables the automatic opening and closing of the bus door after validating the RFID card.

Explanation- The relay module receives a low-voltage control signal (3.3V/5V) from the ESP32 and uses an internal electromagnetic coil to switch the high-power circuit connected to the solenoid lock. When a valid RFID card is scanned, the ESP32 sends a HIGH signal to the relay, turning it ON and energizing the solenoid lock to open the door. After a short delay, the relay turns OFF, locking the door again. This ensures safe, isolated, and automated door control in the bus system.

Specifications-

- Operating Voltage: 5V DC
- Control Signal: 3.3V or 5V logic compatible
- Current Rating: 10A @ 250V AC or 10A @ 30V DC
- Trigger Type: Low-level or high-level (depending on module)
- Electrical Isolation: Optocoupler-based (in most modules)
- Relay Type: SPDT (Single Pole Double Throw)

6.4 RC522 RFID module

Figure 6.4.1 shows the RC522 RFID Module and Interface shown in figure 6.4.2



Figure 6.4.1 RC522 RFID Module

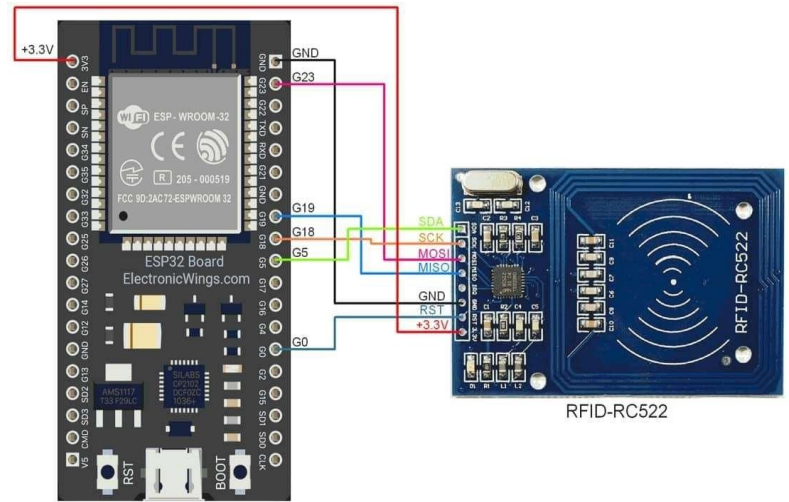


Figure 6.4.2 RC522 RFID Module Interface

Role- The RFID RC522 module is used to authenticate students when they board the school bus. Each student carries an RFID card, and the module reads the card's unique ID and sends it to the ESP32 for validation. Only authorized students are allowed access, ensuring secure entry.

Explanation - The RC522 is a 13.56 MHz RFID reader based on the MFRC522 chipset. It communicates with the ESP32 using the SPI interface, enabling fast and reliable reading of RFID tags. When a student taps their card, the module detects the RFID signal, extracts the UID, and transfers it to the microcontroller for processing. Its short-range communication ensures that only intentional scans are detected, reducing errors and unauthorized access.

Specifications

- Operating Frequency: 13.56 MHz
- Communication Interface: SPI
- Supply Voltage: 3.3 V
- Reading Distance: 2–5 cm
- Supported Cards: MIFARE 1K, 4K, and compatible tags
- Low Power Consumption: Ideal for portable and embedded systems

6.5 LCD 16X2 Display

Figure 6.5.1 shows the LCD Display 16x2 and Interface shown in figure 6.5.2

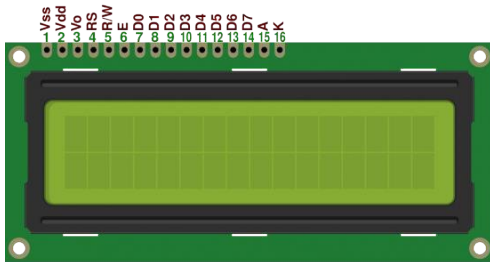


Figure 6.5.1 LCD Display 16x2

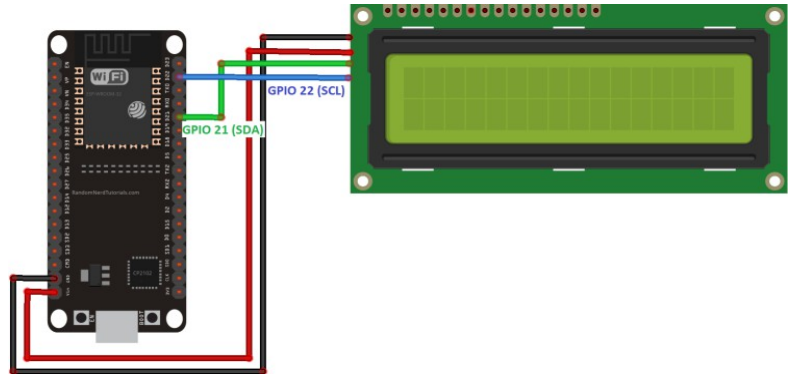


Figure 6.5.2 LCD Display 16x2 Interface

Role-The LCD 16x2 display is used to show real-time system information such as student authentication status, bus status, and safety alerts.

Explanation-The 16x2 LCD (Liquid Crystal Display) can display 16 characters per line on 2 lines, allowing important messages to be shown during system operation. When a student scans their RFID card, the ESP WROOM 32 sends a message to the LCD via I2C communication.

- If the RFID is valid, the LCD displays “Access Granted” along with the student’s ID or name.
- If invalid, it displays “Access Denied.”

Specifications-

- Display Format: 16 Characters × 2 Lines
- Interface Type: 4-bit / 8-bit Parallel or I2C
- Operating Voltage: 5V DC
- Character Font: 5×8 dots
- Communication: I2C (SDA, SCL) or direct GPIO connection to ESP WROOM 32

6.6 Solenoid Lock

Figure 6.6.1 shows the Solenoid Lock and Interface shown in figure 6.6.2



Figure 6.6.1 Solenoid Lock

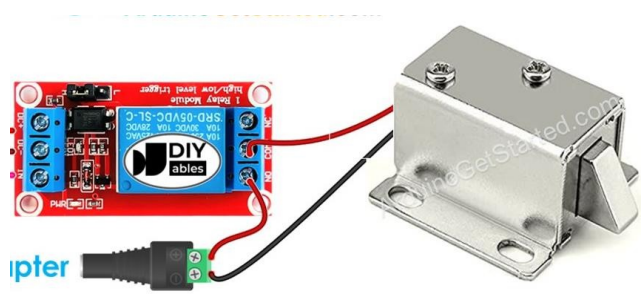


Figure 6.6.2 Solenoid Lock Interface

Role-The solenoid lock acts as the automatic door locking mechanism in the school bus. It ensures that the bus door opens only when an authorized RFID card is scanned, improving student safety and preventing unauthorized entry.

Explanation-A solenoid lock works on the principle of electromagnetism. When the ESP WROOM 32 verifies a valid RFID card, it activates the relay module, which supplies power to the solenoid lock. This energizes the coil inside the lock, creating a magnetic force that pulls or pushes the locking mechanism, allowing the door to open. If the card is invalid, the ESP32 does not energize the solenoid, keeping the door locked.

Specifications-

- Operating Voltage: 12V DC
- Current Consumption: 0.8A – 1A
- Lock Type: Plunger-type electromagnetic solenoid
- Operation Mode: Normally Closed (NC) — unlocks when powered
- Material: Metal body with high mechanical strength
- Control Method: Through relay module controlled by ESP WROOM 32
- Response Time: Fast activation within milliseconds
- Holding Force: Suitable for small to medium-sized doors

6.7 Power Supply Module

Figure 6.6.1 shows the Power Supply Module



Figure 6.7.1 Power Supply Module

Role- The power supply module provides stable and regulated voltage levels required for different components in the project. It powers the ESP32, RFID module, LCD display, GPS module, and relay–solenoid lock arrangement, ensuring reliable operation of the entire system.

Explanation- The 5V–12V power supply module converts an unregulated input (like adapter or battery) into stable DC voltage outputs. In your project, 5V is used for logic-level components such as ESP32, RFID RC522, GPS Neo-6M, LCD, and relay signal inputs, while 12V powers the solenoid lock. This module prevents voltage fluctuations, ensures smooth current flow, and protects sensitive components from damage. It also maintains common grounding across the circuit for stable communication between modules.

Specifications

- Input Voltage: 7–24V DC
- Output Voltages: 5V & 12V regulated
- Output Current: Typically, 1A–2A depending on module type
- Regulator Type: Buck/linear voltage regulator
- Protections: Overload, short-circuit, and reverse polarity (depends on model)
- Terminals: Screw terminals or barrel jack for easy wiring

6.8 Circuit diagram

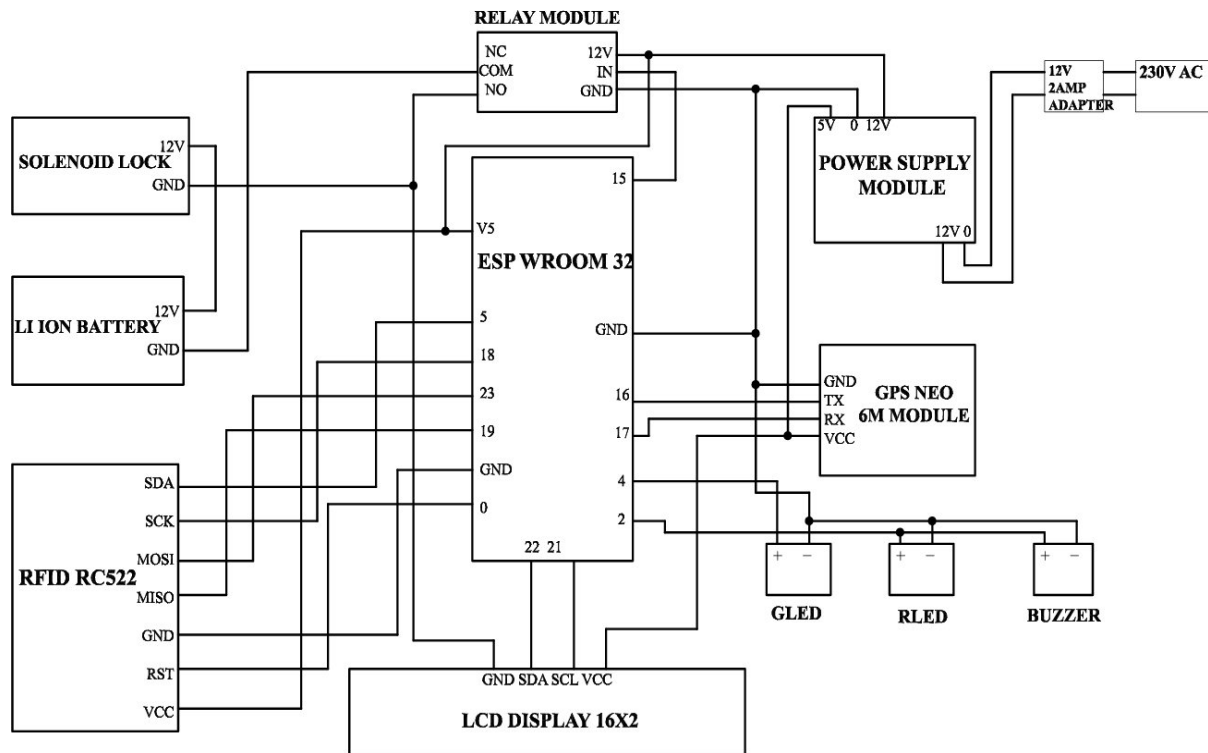


Figure 6.8.1 Circuit Diagram

The circuit integrates multiple modules with the ESP WROOM 32 microcontroller, which acts as the central processing unit for the entire system. The RFID RC522 module is interfaced to the ESP32 using SPI pins to read student ID cards for authentication. The GPS Neo-6M module is connected through UART pins to provide real-time location data. A 16×2 I2C LCD is used to display authentication status and system messages. The relay module is driven by a GPIO pin of the ESP32 and controls the solenoid door lock, enabling it to open only when a valid RFID card is detected. A buzzer and LED indicators are connected to separate GPIO pins to give audio-visual alerts during door operation and invalid scans. The circuit is powered through a 12V power supply module, the Li-ion battery is separately charged and provides regulated voltage to the ESP32 and peripherals. Overall, the circuit ensures secure access control, real-time monitoring, and efficient power management for the IoT-based school bus system. Figure 6.8.1 shows the circuit diagram of the project.

CHAPTER 7

SOFTWARE DESCRIPTION

After understanding the hardware components and their interconnections, suitable software tools are necessary to control and coordinate their operation. This chapter discusses the software platforms and tools used to program and monitor the system. It includes the Arduino IDE for coding and the Arduino IoT Cloud for real-time monitoring. The software ensures proper communication between hardware modules and enables reliable system performance.

7.1 Arduino IDE

Arduino uno for programming Arduino microcontroller. Arduino uno can be implemented within Windows, Mac and Linux operating systems. Most of its components are written in c language for easy editing and compiling. Users can modify internal layouts and schematics when required.

Integrated Development Environment “IDE” For Arduino

Arduino IDE is an open-source software that is mainly used for writing and compiling the code into the Arduino Module.

It is an official Arduino software, making code compilation too easy that even a common person with no prior technical knowledge can get their feet wet with the learning process. It is easily available for operating systems like MAC, Windows, Linux and runs on the Java Platform that comes with inbuilt functions and commands that play a vital role for debugging, editing and compiling the code in the environment.

Tools

Auto Format This formats your code nicely: i.e., indents it so that opening and closing curly braces line up, and that the statements inside curly braces are indented more. **Archive Sketch** Archives a copy of the current sketch in .zip format. The archive is placed in the same directory as the sketch.

Fix Encoding & Reload Fixes possible discrepancies between the editor char map encoding and other operating systems char maps. **Serial Monitor** Opens the serial monitor window and initiates the exchange of data with any connected board on the currently selected Port. This usually resets the board, if the board supports Reset over serial port opening.

Board Select the board that you're using. See below for descriptions of the various boards. This menu contains all the serial devices (real or virtual) on your machine. It should automatically refresh every time you open the top-level tools menu. **Programmer** For selecting a hardware programmer when programming a board or chip and not using the onboard USB serial connection. Normally you won't need this, but if you're burning a bootloader to a new microcontroller, you will use this. **Burn Bootloader** The items in this menu allow you to burn a bootloader onto the microcontroller on an Arduino board. This is not required for normal use of an Arduino board but is useful if you purchase a new ATmega microcontroller (which normally come without a bootloader). Ensure that you've selected the correct board from the Boards menu before burning the bootloader on the target board. This command also set the right fuses.

Writing Sketches

Programs written using Arduino Software (IDE) are called sketches. These sketches are written in the text editor and are saved with the file extension. `.ino`. The editor has features for cutting/pasting and for searching/replacing text. The message area gives feedback while saving and exporting and also displays errors. The console displays text output by the Arduino Software (IDE), including complete error messages and other information. The bottom righthand corner of the window displays the configured board and serial port. The toolbar buttons allow you to verify and upload programs, create, open, and save sketches, and open the serial monitor. Arduino IDE is used to control all the input and output devices that are connected to the Arduino uno with programming.

7.2 Arduino IoT Remote App

The Arduino IoT Remote App (available for Android and iOS) is an extension of the Arduino Cloud that allows users to monitor and control IoT devices directly from a smartphone. It connects securely to the cloud and displays data stored in it.

Role in this Project:

- The app displays the live GPS location of the school bus using data from the GPS Neo-6M module.
- It shows RFID authentication logs such as student entry times and card verification status.
- Parents and school authorities can access this data remotely from any location.
- Provides real-time updates on the bus route, ensuring safety and transparency.

Working Process:

- The ESP WROOM 32 reads RFID and GPS data.

- This data is transmitted to the Arduino Cloud via Wi-Fi.
- The Arduino IoT Remote App fetches the data from the cloud and displays it on a user-friendly dashboard.
- Users can monitor live bus location and student status in real time.

Advantages:

- Provides a user-friendly interface for remote monitoring.
- Works seamlessly with Wi-Fi-connected ESP boards.

7.3 Flow Chart

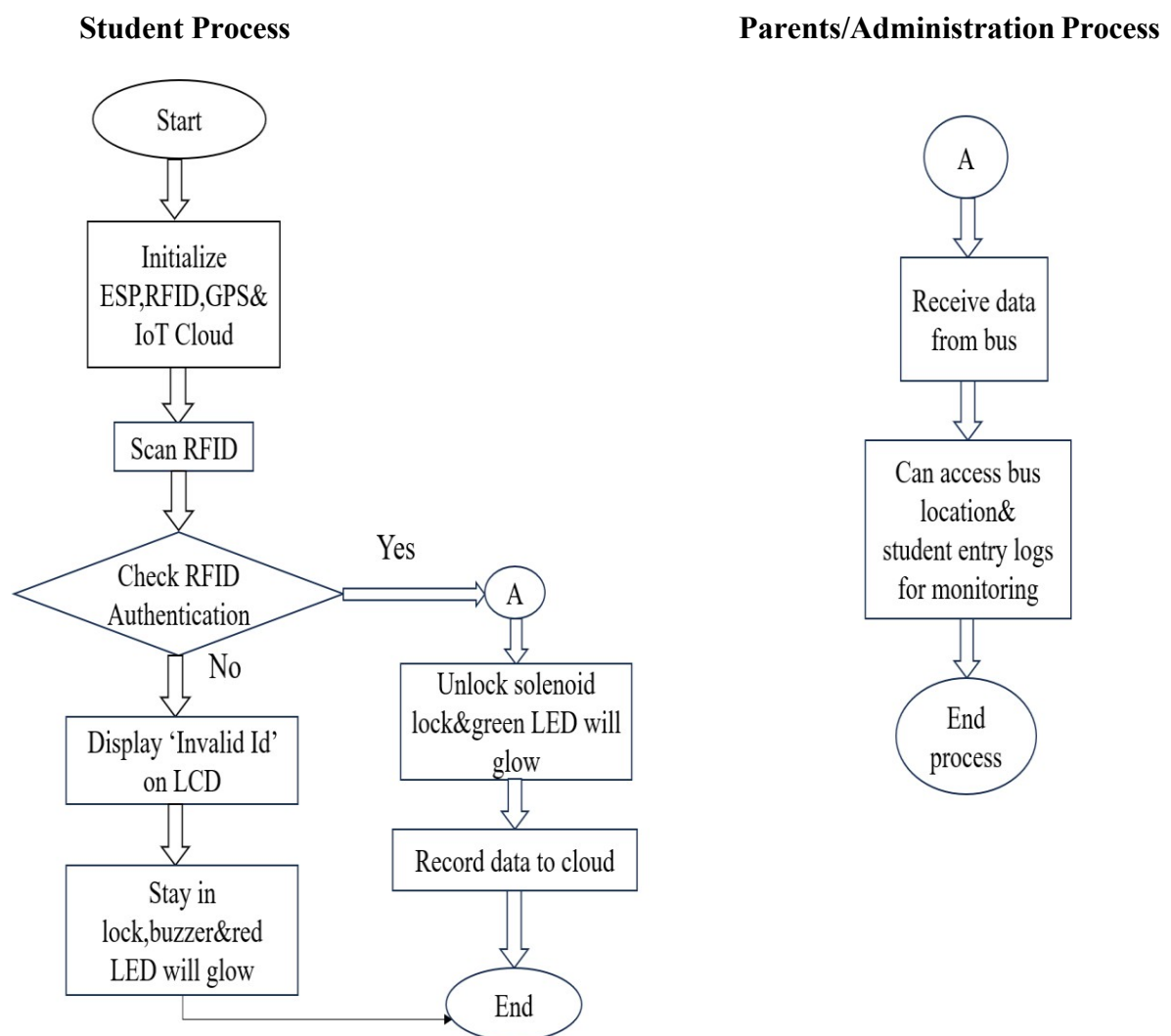


Figure 7.3.1 Flow Chart

The overall flow chart of the proposed IoT-Based School Bus Management System describes the sequence of operations carried out by both the student side process and the administration/parent

monitoring process. The system begins by initializing all essential hardware modules, including the ESP32 microcontroller, RFID scanner, GPS module, and the IoT Cloud connectivity. Once the system is powered on and initialized, it continuously waits for a student to scan their RFID card at the bus entry point.

In the Student Process, the first active step occurs when a student presents their RFID card to the scanner. The ESP32 reads the unique ID and verifies whether it matches any authorized ID stored in the system database. If the scanned card is found to be invalid, the system displays “Invalid ID” on the LCD screen, activates the red LED and buzzer alert to indicate denial of access, and keeps the solenoid lock in locked state. This ensures that unauthorized students cannot board the bus and enhances overall safety.

If the RFID authentication is successful, the control shifts to the next stage (point A in the flow chart). Here, the system triggers the relay module, which in turn activates the solenoid lock, temporarily unlocking the door to allow the authorized student to enter. Simultaneously, the green LED glows, indicating successful verification. Once the student boards the bus, the system automatically records the authentication event and updates the IoT Cloud database. This cloud entry contains essential information such as the verified student ID, date, time, and authentication status. After recording the information, the student-side process ends and the system returns to its initial state, waiting for the next RFID scan. This continuous loop ensures a smooth boarding process for every student.

The Parents/Administration Process works in parallel with the student process. Once the data is uploaded to the cloud, the administration module retrieves the updated information (represented by section A on the right side of the chart). Through the IoT Cloud interface or mobile application, school authorities and parents can access the live bus location along with the student entry records. This enables real-time monitoring without manual intervention. Parents can view whether their child has boarded the bus, while school authorities can supervise the functioning of the transportation system efficiently.

The process concludes once the cloud data has been viewed, but the system remains active for subsequent updates as more students scan their RFID cards during pick-up or drop-off times. This structure ensures a reliable and automated flow of information between the bus system and the monitoring units.

Overall, the flow chart effectively highlights how the proposed system replaces manual verification with a secure, automated, and IoT-enabled mechanism. Figure 7.3.1 shows the flow chart of the project.

CHAPTER 8

RESULTS AND DISCUSSION

After completing the hardware setup and software implementation, the system was tested under various conditions. This chapter presents the results obtained from the experimental evaluation of the project. The observed outputs are analyzed and discussed to verify system performance. The discussion demonstrates how effectively the system meets the defined objectives.

8.1 System Functionality Results

8.1.1 RFID Authentication Module

Figure 8.1.1 shows the valid id (lock opened with green led) and figure 8.1.2 shows the Invalid Id (lock stays with red led & buzzer)

- The RFID-RC522 successfully detected and read the unique IDs of student RFID cards.
- Valid IDs triggered the unlocking of the solenoid lock and glowing of the green LED, confirming successful authentication.
- Invalid IDs produced an “Invalid ID” message on the LCD, kept the lock closed, and activated the red LED + buzzer alert.
- The module showed fast response time (less than 1 second), indicating reliable short-range authentication suitable for school bus entry control.

8.2.2 Solenoid Lock & Relay Operation

- The solenoid lock operated correctly through the 1-channel relay module.
- For valid authentication, the ESP32 activated the relay, which allowed 12V power to energize the solenoid lock and open the door.
- When an invalid card was scanned, the relay remained OFF, ensuring the door stayed locked.
- The mechanism proved effective in preventing unauthorized boarding, enhancing bus safety.

8.2.3 GPS Neo-6M Tracking Module

- The GPS module consistently provided real-time coordinates with good accuracy under open-sky conditions.
- These coordinates were sent to the Arduino IoT Cloud through ESP32's Wi-Fi.

- The mobile IoT dashboard displayed live bus location updates, allowing parents and administrators to track the bus movement at any time.

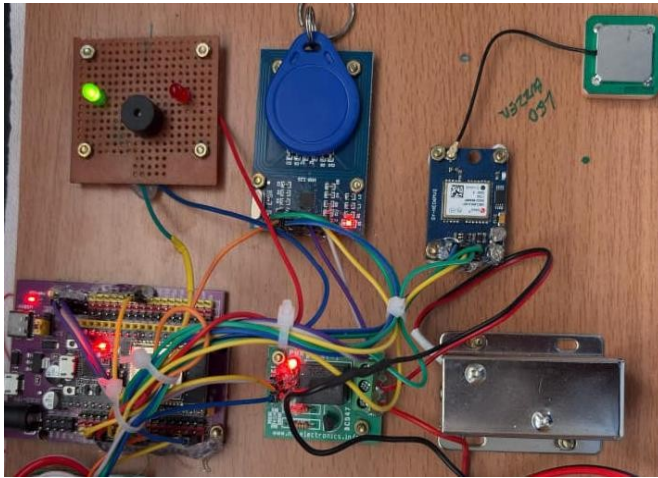


Figure 8.1.1 Valid id (lock opened with green led)

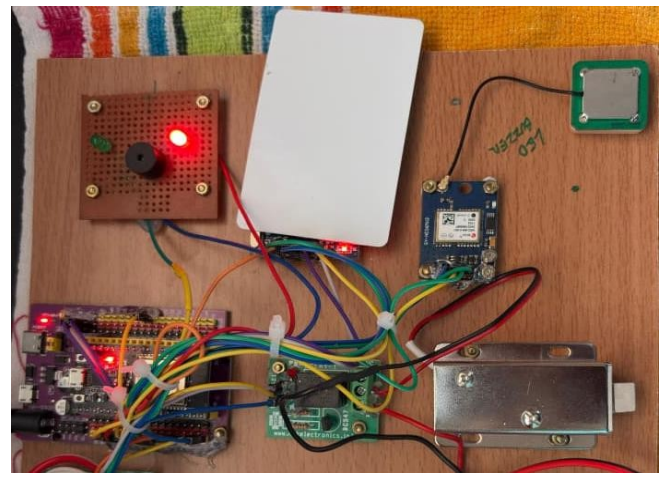


Figure 8.1.2 Invalid Id (lock stays with red led & buzzer)

8.2.4 LCD 16x2 Module

Figure 8.1.3 shows the Lcd outputs

- The LCD 16x2 module displayed all system messages clearly, including “Valid ID”, “Invalid ID”, door closing status & welcome statement.
- The display acted as a real-time feedback interface for students and the driver, showing authentication results immediately after the RFID card scan.
- The module performed efficiently with the ESP32 via the I²C interface, requiring minimal wiring while maintaining stable output throughout testing.



Figure 8.1.3 Lcd outputs

8.2 Cloud Monitoring Module

8.2.1 Arduino IoT Remote App

Figure 8.1.1 shows the data stored in cloud

- The Arduino IoT Remote App allowed real-time monitoring of the school bus location by receiving continuous GPS updates from the ESP32 through the Arduino IoT Cloud.
- The app displayed essential system parameters—including bus coordinates, authentication logs—ensuring transparent communication between bus system and users.
- Parents and school authorities could instantly view the bus movement on a mobile dashboard, improving safety supervision and reducing uncertainty during transit.
- The app provided reliable cloud synchronization, enabling smooth data updates even when multiple variables (location, lock status, scan logs) were transmitted simultaneously.

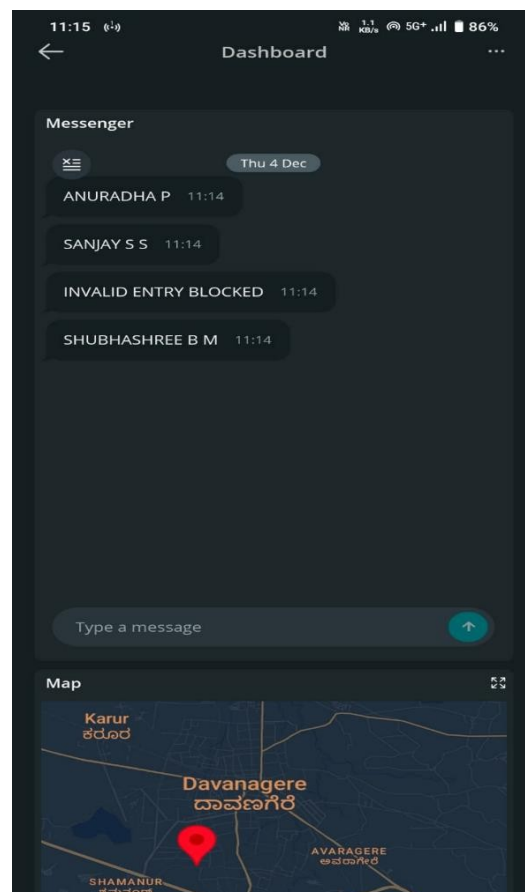


Figure 8.2.1 Data stored in Cloud

8.3 Key Insights

Figure 8.3.1 shows the complete project model

- The integration of RFID, GPS, and IoT Cloud provides a complete safety ecosystem for school buses.
- The prototype demonstrated that unauthorized entry can be efficiently prevented, addressing a major concern in school transportation.
- Real-time location availability enhances transparency and reduces anxiety among parents and school administrators.
- The use of ESP32 enables simultaneous handling of authentication, GPS processing, and Wi-Fi-based cloud updates, proving its suitability for IoT transportation systems.
- The system showed excellent scalability — more students, multiple buses, and additional sensors can be added easily in future versions.
- Overall, the system proved to be efficient, reliable, responsive, and technically feasible for adoption in real-world school environments.

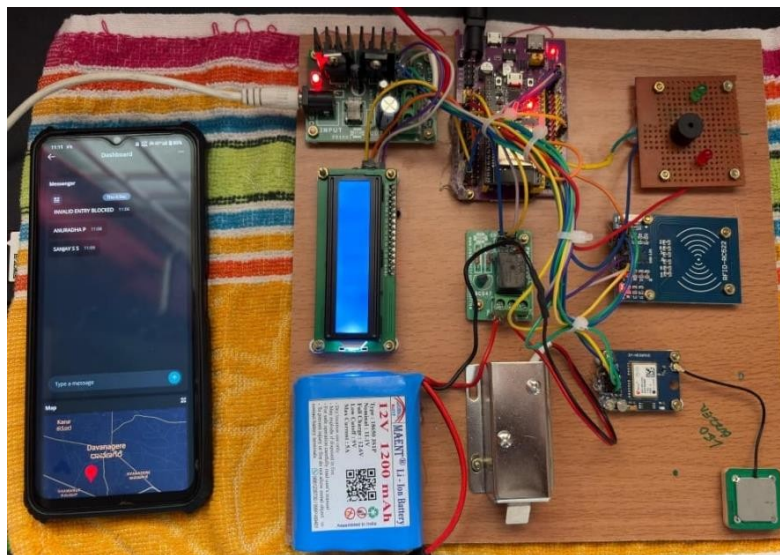


Figure 8.3.1 Project Model

ADVANTAGES

Main advantages of this project are:

- **Enhanced Student Safety**

The system ensures that only authorized students can board the bus through RFID authentication, preventing unauthorized access and enhancing overall safety.

- **Real-Time Monitoring**

Parents and school authorities can view the bus's live location and student boarding status through the Arduino IoT Cloud and IoT Remote App.

- **Instant Alerts and Notifications**

The system provides immediate feedback through the LCD display, buzzer, and LED indicators, ensuring quick response to any irregular situation.

- **Data Storage and Accessibility**

All data such as bus location, student entry logs, and system status are stored in the IoT cloud, allowing easy access and analysis anytime, anywhere.

- **Energy Efficient and Reliable Operation**

The use of a Li-ion battery backup ensures uninterrupted power supply and efficient energy usage, making the system reliable even during power failure.

- **Ease of Integration and Scalability**

The system can be easily integrated into existing school transport infrastructure and scaled for multiple buses with minimal changes.

- **Transparency and Parental Confidence**

Real-time visibility of bus location and student safety status gives parents peace of mind and builds trust in school transportation.

LIMITATIONS

- **Internet Dependency**

The system relies on a stable internet connection for real-time data upload to the Arduino IoT Cloud. Poor network coverage can cause delayed updates.

- **RFID Card Misuse**

Since the system lacks biometric or facial recognition, anyone carrying a student's RFID card could misuse it to gain unauthorized access.

- **Power and Maintenance**

The system requires proper battery maintenance and a continuous power supply for reliable performance.

- **Initial Cost and Scalability**

Setting up multiple buses requires more devices and cloud subscriptions, increasing initial implementation cost.

APPLICATIONS

- **Student Safety and Security**

The system ensures that only authorized students can board the bus through RFID authentication and automatic door control, minimizing the risk of unauthorized access or missing students.

- **Real-Time Tracking and Monitoring**

Using the GPS module and IoT cloud integration, school authorities and parents can monitor the live location of the bus through the Arduino IoT Remote App, enhancing visibility and safety.

- **Parental Notification and Transparency**

Parents can track when their child boards or leaves the bus, promoting transparency and peace of mind.

- **Efficient School Transport Management**

School administrators can analyze route efficiency, monitor delays, and optimize bus schedules using collected GPS and RFID data.

- **Scalable Smart Transportation System**

The same concept can be extended to other institutions, company transport, or public buses for smart and automated fleet management.

CONCLUSION

The IoT-Based School Bus Management System Using RFID successfully enhances student safety and monitoring through secure RFID authentication, GPS-based location tracking, and cloud connectivity. By integrating ESP WROOM 32 with RFID RC522, GPS Neo-6M, solenoid lock, and IoT Cloud, the system ensures that only authorized students can access the bus while providing real-time visibility of bus movement to parents and school authorities. The use of a solenoid lock, LED indicators, LCD display, and buzzer further strengthens the safety mechanism, ensuring immediate feedback during identification. Overall, the system delivers a reliable and efficient approach to improving school transportation safety and transparency.

This project demonstrates the practical application of IoT in addressing real-world safety concerns in school transportation. The integration of cloud-based monitoring allows authenticated entry logs and live bus location to be accessed remotely, enabling proactive decision-making during emergencies or delays. The modular design also offers scalability, making it adaptable for colleges, corporate transport, and public vehicles. With further enhancements such as biometric verification or camera-based identification, the system can be expanded into a fully secure and intelligent transportation ecosystem.

FUTURE SCOPE

- **Integration of Face Recognition / Biometrics-** Adding a camera-based or fingerprint-based verification system will reduce misuse of RFID cards and improve authentication accuracy.
- **Offline Data Storage & Syncing-** Implementing local memory storage will allow the system to record data even without internet connectivity and sync it once the network is restored.
- **Geo-fencing & Auto-Alerts-** The system can send automatic alerts when the bus enters or leaves predefined areas such as school, bus stop, or restricted zones.
- **Fuel and Driver Behavior Monitoring-** Sensors can be added to track fuel consumption, engine condition, speed, and driver behavior to improve fleet management.
- **Adoption of LoRa/GSM Hybrid Network-** For areas with poor Wi-Fi coverage, using long-range LoRa or GSM backup networks ensures uninterrupted data transmission.
- **Emergency Detection System-** Additional sensors (vibration, flame, temperature) can be used to detect accidents or bus abnormalities and automatically send alerts.

REFERENCES

- [1] R. R. Das, D. R. Sona, and D. Deepika, “IoT-Based School Bus and Student Monitoring System using RFID and GSM Technologies,” *International Journal of Innovative Science and Applied Engineering (IJISAE)*, vol. 7, no. 2, pp. 1556–1562, Feb. 2024.
- [2] N. Shaji, T. Salih, and A. Jose, “Digitalization of College Bus Transportation System,” *International Journal of Engineering Research & Technology (IJERT)*, vol. 11, no. 4, pp. 1–5, Apr. 2023.
- [3] B. Nithin, M. Srujan, S. Amith, and K. Akhilesh, “College Bus Security and Student Monitoring System,” *Journal of Emerging Technologies and Innovative Research (JETIR)*, vol. 9, no. 7, pp. 110–115, July 2022.
- [4] M. Malleswari and K. Srujana, “RFID Based College Bus Management System,” *International Research Journal of Engineering and Technology (IRJET)*, vol. 5, no. 3, pp. 2360–2365, Mar. 2018.
- [5] P. A. Kamble and R. Vatti, “Bus Tracking and Monitoring Using RFID,” *International Journal of Advanced Research in Computer Engineering & Technology (IJARCET)*, vol. 6, no. 7, pp. 1034–1040, July 2017.

APPENDIX-A

DATASHEETS

Sl. No	Component	Specification	Function in Project
1.	ESP WROOM 32	- Dual-core 32-bit MCU (Tensilica LX6) - Wi-Fi + Bluetooth - Operating Voltage: 3.3V - GPIO, ADC, PWM support - Flash memory: 4–16 MB	Acts as the main controller for the project; processes RFID & GPS data, controls solenoid lock & relay, and sends data to Arduino IoT Cloud for monitoring.
2.	RFID Scanner (RC522)	- Frequency: 13.56 MHz - Interface: SPI - Operating Voltage: 3.3V - Read Range: 2–5 cm	Used for student authentication. Reads RFID card UID and sends data to ESP32 for verification.
3.	GPS Module (Neo-6M)	- Position Accuracy: 2.5m - Operating Voltage: 3.3–5V - Communication: UART (TX/RX) - Baud Rate: 9600	Provides live location tracking of the school bus and transmits coordinates to ESP32 for cloud update.
4.	LCD Display (16x2, I2C)	- Display: 16 characters × 2 lines - Interface: I2C (SDA, SCL) - Operating Voltage: 5V - Backlight: LED	Displays real-time information such as student entry status (Valid/Invalid), lock status, and system messages.
5.	Solenoid Lock (12V)	- Operating Voltage: 12V - Type: Push/Pull (Electromagnetic) - Current: 0.5–1A	Locks/Unlocks the bus door automatically based on valid RFID authentication. Controlled via relay.
6.	Relay Module (1-Channel)	- Trigger Voltage: 3.3–5V - Load Voltage: AC/DC up to 230V - Contact Rating: 10A	Used as a switching device to operate the 12V solenoid lock safely using a low-power signal from the ESP32.
7.	LED Indicators (Red/Green)	- Operating Voltage: 2–3V - Current: 20mA	Green LED → Valid RFID & door unlocked Red LED → Invalid RFID or error status
8.	Buzzer	- Operating Voltage: 3–5V - Type: Active buzzer	Provides audio alerts for invalid RFID scans and system warnings.
9.	Li-ion Battery (12V)	- Output Voltage: 12V - Capacity: 2200–4400mAh - Rechargeable	Acts as a backup power source for solenoid lock.
10.	5–12V Power Supply Module	- Input: 230V AC - Output: 12V / 5V DC - On-board voltage regulators	Converts mains AC supply into stable 5V and 12V outputs required for ESP32, RFID, LCD, GPS, and solenoid lock.

ESP WROOM 32

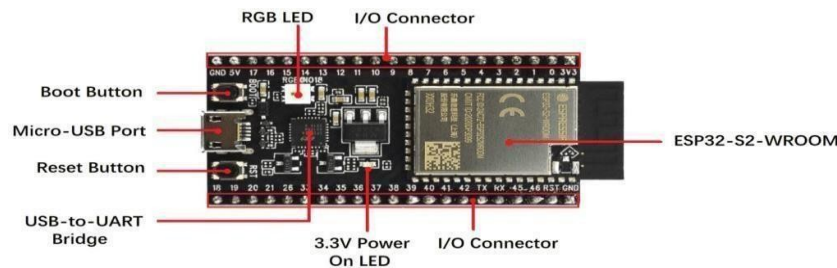


Figure ESP 32 Specifications

Here is a specification chart for the **ESP-WROOM-32** module

Specification	Details
Microcontroller	ESP32-D0WD (varies by model)
Architecture	Xtensa® 32-bit LX6 dual-core
Clock Speed	Up to 240 MHz
Flash Memory	Typically 4 MB (can be 8 MB or 16 MB)
SRAM	520 KB (including 8 KB RTC fast memory, 8 KB RTC slow memory)
Wireless Connectivity	Wi-Fi 802.11 b/g/n (2.4 GHz), Bluetooth 4.2 BR/EDR & BLE
GPIO Pins	Up to 34 (multipurpose)
ADC Channels	18 channels (but only 15 usable)
DAC Channels	2 (8-bit resolution)
UART	3
SPI	4
I2C	2
PWM	Available on most GPIOs
Operating Voltage	3.3V
Power Consumption	Active: ~160 mA, Deep Sleep: ~10 µA
Operating Temperature	-40°C to 85°C
Dimensions	18 mm × 25.5 mm × 3.1 mm

ESP32-WROOM Pinout Description

The ESP32-WROOM module has multiple GPIO pins supporting various functions like ADC, DAC, PWM, I2C, SPI, UART, and more. Below is the standard pinout for ESP32-WROOM:

Power & Ground Pins:

1. 3V3 – 3.3V power supply input
2. GND – Ground

GPIO & Special Function Pins:

- GPIO0 – Boot mode selection (must be pulled LOW for flashing)
- GPIO1 (TX0), GPIO3 (RX0) – Default UART0 TX and RX pins
- GPIO2 – Strapping pin, used for boot mode
- GPIO4, GPIO5, GPIO12, GPIO13, GPIO14, GPIO15, GPIO16, GPIO17 – General-purpose I/O pins
- GPIO6 to GPIO11 – Connected to internal SPI flash (not recommended for external use)
- GPIO18 (SCK), GPIO19 (MISO), GPIO23 (MOSI), GPIO5 (SS) – SPI interface pins
- GPIO21 (SDA), GPIO22 (SCL) – I2C communication pins
- GPIO32 to GPIO39 – ADC (Analog to Digital Converter) capable pins
- GPIO25, GPIO26 – DAC (Digital to Analog Converter) output pins
- GPIO34, GPIO35, GPIO36 (VP), GPIO39 (VN) – Input-only ADC pins (cannot be used as outputs)

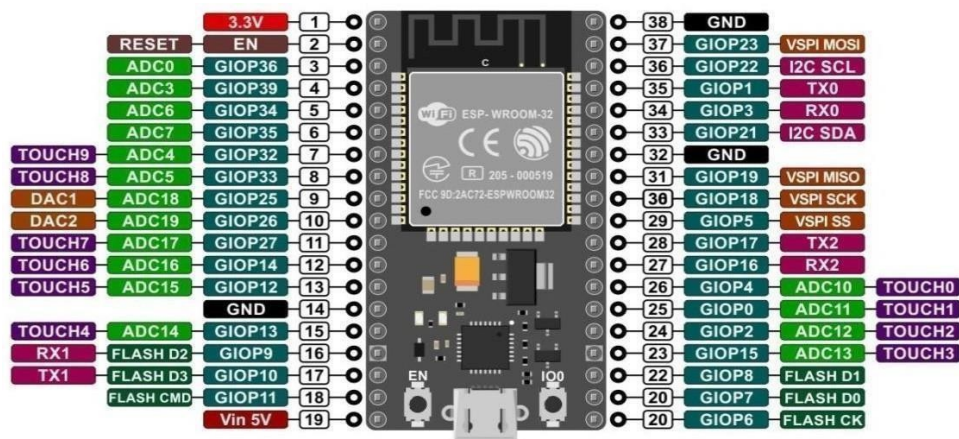


Figure EPR WROOM 32 Pinout Diagram

APPENDIX-B

SOURCE CODE

```
#include <TinyGPS++.h>
#define RXD2 16
//EVEN THOUGH 16 DEFINED CONNECT RX TO P17
#define TXD2 17
//EVEN THOUGH 17 DEFINED CONNECT TX TO P16
#define GPS_BAUD 9600
TinyGPSPlus gps;
HardwareSerial gpsSerial(2);

//PROJECT
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
LiquidCrystal_I2C lcd(0x27, 16, 2);
#include <SPI.h>
#include <MFRC522.h> //USE BELOW LINK LIBRARY
OTHERS WONT WORK
//https://github.com/makerspaceleiden/rfid

#define SS_PIN 5 // SDA
#define RST_PIN 32 // RST
// SCK=18, MOSI=23, MISO=19, (ESP32 default)
#define RELAY_PIN 15
#define RLED 2
#define GLED 4

MFRC522 rfid(SS_PIN, RST_PIN);

////READ THE TAG VALUE AND STORE HERE
byte authorizedUID1[4] = {0x89, 0x53, 0xE6, 0x00};
byte authorizedUID2[4] = {0x27, 0x81, 0xE7, 0x00};
byte authorizedUID3[4] = {0x19, 0x7F, 0x17, 0x05};
byte authorizedUID4[4] = {0x9B, 0xEE, 0x14, 0x05};

void setup()
{
  Serial.begin(9600);
  Serial.begin(115200);
  gpsSerial.begin(GPS_BAUD, SERIAL_8N1, RXD2, TXD2);
  Serial.println("Serial 2 started at 9600 baud rate");
  SPI.begin(); // init SPI bus
  rfid.PCD_Init(); // init MFRC522
  pinMode(RELAY_PIN, OUTPUT); // initialize pin as an output.
```

```
//digitalWrite(RELAY_PIN, LOW); // deactivate the relay
pinMode(RLED, OUTPUT);
pinMode(GLED, OUTPUT);
Serial.println("Tap an RFID/NFC tag on the RFID-RC522 reader");
lcd.begin();
lcd.backlight();
lcd.setCursor(0, 0);
lcd.print("BUS RESERVATION ");
lcd.setCursor(0, 1);
lcd.print("BIET COLLEGE DVG");
delay(5000);
lcd.clear();
}
```

```
void loop() {
  if (rfid.PICC_IsNewCardPresent()) { // new tag is available
    if (rfid.PICC_ReadCardSerial()) { // NUID has been readed
      MFRC522::PICC_Type piccType =
        rfid.PICC_GetType(rfid.uid.sak);
```

```
      ///TAG1
      if (rfid.uid.uidByte[0] == authorizedUID1[0] &&
          rfid.uid.uidByte[1] == authorizedUID1[1] &&
          rfid.uid.uidByte[2] == authorizedUID1[2] &&
          rfid.uid.uidByte[3] == authorizedUID1[3] )
      { Serial.println("ANURADHA P");
        lcd.setCursor(0, 0);
        lcd.print("          ");
        lcd.setCursor(0, 0);
        lcd.print("ANURADHA P");
        lcd.setCursor(0, 1);
        lcd.print("VALID-ID..ENTER..");
        open();
      }
}
```

```
      ///TAG2
      else
      if (rfid.uid.uidByte[0] == authorizedUID2[0] &&
          rfid.uid.uidByte[1] == authorizedUID2[1] &&
          rfid.uid.uidByte[2] == authorizedUID2[2] &&
          rfid.uid.uidByte[3] == authorizedUID2[3] )
      { Serial.println("SANJAY S S");
        lcd.setCursor(0, 0);
        lcd.print("          ");
        lcd.setCursor(0, 0);
        lcd.print("SANJAY S S");
        lcd.setCursor(0, 1);
```

```
    lcd.print("VALID-ID..ENTER..");
    open();
}

///  
TAG3
else
if (rfid.uid.uidByte[0] == authorizedUID3[0] &&
    rfid.uid.uidByte[1] == authorizedUID3[1] &&
    rfid.uid.uidByte[2] == authorizedUID3[2] &&
    rfid.uid.uidByte[3] == authorizedUID3[3] )
{ Serial.println("SHUBHASHREE B M");
  lcd.setCursor(0, 0);
  lcd.print("          ");
  lcd.setCursor(0, 0);
  lcd.print("SHUBHASHREE B M");
  lcd.setCursor(0, 1);
  lcd.print("VALID-ID..ENTER..");
  open();
}

///  
TAG4
else
if (rfid.uid.uidByte[0] == authorizedUID4[0] &&
    rfid.uid.uidByte[1] == authorizedUID4[1] &&
    rfid.uid.uidByte[2] == authorizedUID4[2] &&
    rfid.uid.uidByte[3] == authorizedUID4[3] )
{ Serial.println("DEEPA N H");
  Serial.print("Unauthorized");
  lcd.setCursor(0, 0);
  lcd.print("          ");
  lcd.setCursor(0, 0);
  lcd.print("DEEPA N H");
  lcd.setCursor(0, 1);
  lcd.print("INVALID!!!..STOP!");
  alert();
}
else
{
  Serial.print("Not Registered Tag with UID:");
  lcd.setCursor(0, 1);
  lcd.print("NOT..REGISTERED!!");
  alert();
  for (int i = 0; i < rfid.uid.size; i++)
  { Serial.print(rfid.uid.uidByte[i] < 0x10 ? " 0" : " ");
    Serial.print(rfid.uid.uidByte[i], HEX);
  }
  Serial.println();
}
```

```
}

rfid.PICC_HaltA(); // halt PICC
rfid.PCD_StopCrypto1(); // stop encryption on PCD
}
}

////GPS
unsigned long start = millis();
while (millis() - start < 1000)
{ while (gpsSerial.available() > 0)
{
    gps.encode(gpsSerial.read());
}
if (gps.location.isUpdated()) {
    // Serial.print("LAT: ");
    // Serial.println(gps.location.lat(), 6);
    // Serial.print("LONG: ");
    // Serial.println(gps.location.lng(), 6);
    // location = {gps.location.lat() , gps.location.lng()};
    // ArduinoCloud.update();
    // Serial.print("SPEED (km/h) = ");
    // Serial.println(gps.speed.kmph());
    // Serial.print("ALT (min)= ");
    // Serial.println(gps.altitude.meters());
    // Serial.print("HDOP = ");
    // Serial.println(gps.hdop.value() / 100.0);
    // Serial.print("Satellites = ");
    // Serial.println(gps.satellites.value());
    // Serial.print("Time in UTC: ");
    // Serial.println(String(gps.date.year()) + "/" +
String(gps.date.month()) + "/" + String(gps.date.day()) + "," +
String(gps.time.hour()) + ":" + String(gps.time.minute()) + ":" +
String(gps.time.second()));
    // Serial.println("");
}
}
}

void open()
{
    digitalWrite(RELAY_PIN, HIGH);
    digitalWrite(GLED,HIGH);
    delay(5000);
    digitalWrite(GLED,LOW);
    digitalWrite(RLED,HIGH);
    lcd.setCursor(0, 1);
```

```
lcd.print("DOORS...CLOSING..");
```

```
    delay(5000);
    digitalWrite(RLED,LOW);
    digitalWrite(RELAY_PIN, LOW);
    lcd.clear();
}

void alert()
{
    digitalWrite(GLED,LOW);
    digitalWrite(RLED,HIGH);
    delay(500);
    digitalWrite(RLED,LOW);
    delay(500);
    digitalWrite(RLED,HIGH);
    delay(500);
    digitalWrite(RLED,LOW);
    delay(500);
    digitalWrite(RLED,HIGH);
    delay(500);
    digitalWrite(RLED,LOW);
    delay(5000);
    lcd.clear();
}
```

COURSE OUTCOMES

On completion of this course, students are able to:

1. Get exposed to literature available for research challenges in interdisciplinary fields.
2. Get an insight of real-world problems, challenges and related engineering issues to broader societal context.
3. Implement their idea using modern tools & new technologies.
4. Demonstrate professionalism with ethics, individual & team management and effective communication skills to comprehend & to write reports.

COST ESTIMATION

Sl.No	Component	Qty	Total Price
1.	ESP WROOM 32	1	Rs 350
2.	RFID RC 522	1	Rs 90
3.	GPS Neo 6m	1	Rs 250
4.	LCD 16x2	1	Rs 150
5.	Solenoid Lock	1	Rs 310
6.	Relay 1 channel	1	Rs 50
7.	12V Li ion Battery	1	Rs 500
8.	Power supply module	1	Rs 250
9.	Led	2	Rs 5
10.	Buzzer	1	Rs 30
11.	Miscellaneous (wires, breadboard, connectors, casing)	-	Rs 500
	Total		Rs 2485

CONTACT DETAILS

Name: Anuradha P USN No:4BD22EC011 Email id: aanu74024@gmail.com Mobile No: 9986820622 Address: KTJNagar 13th cross 3rd main Davangere,577002	
Name: Sanjay S Sheshagiri USN No:4BD22EC079 Email id: sanjaysheshagiri9@gmail.com Mobile No: 9353823419 Address: Cholamardeshwara nagar 1st main 1st cross, Ranebennur-581115	
Name: Shubhashree BM USN No:4BD22EC121 Email id: shubhabm927@gmail.com Mobile No: 8792196839 Address: Vijaya nagara badavane 1st main 6th cross Davanagere,577001	